

JECRC UNIVERSITY, JAIPUR
Session: Jan-June 2026
Subject: Mathematics-II (DMA025A)
B. Tech. II Semester I Year 2025-26

Tutorial Sheet-2

(M.M: 64)

UNIT-1: Matrices

CO1: The students will learn the essential tool of matrices and linear algebra in a comprehensive manner.

PART – A

(5*2=10)

A1: Define linear dependent and linear independent of vectors.

A2: Find the eigen values of the matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$.

Ans.: 1, 2, 3

A3: Define orthogonal matrix.

A4: Define Cayley Hamilton Theorem.

A5: Define model matrix.

PART – B

(3*7=21)

B1: Are the vectors $x = (1, 3, 4, 2)$, $y = (3, -5, 2, 2)$ and $z = (2, -1, 3, 2)$ linearly dependent? If so express one of these as a linear combination of others.

B2: Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$.

Ans.: 0, 3, 15

B3: Verify Cayley Hamilton Theorem of the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$.

PART – C

(3*11=33)

C1: Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$.

Ans.: 2, 2, 8

C2: Verify Cayley Hamilton Theorem and hence find A^{-1} of the matrix $A = \begin{bmatrix} 1 & -2 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & 2 \end{bmatrix}$.

Ans.: $A^{-1} = \frac{1}{9} \begin{bmatrix} 7 & -3 & -15 \\ -2 & -3 & -1 \\ 9 & 16 & 34 \end{bmatrix}$.

C3: Diagonalize the matrix $A = \begin{bmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{bmatrix}$.

Ans.: Eigen values are 1, 3 and -7.
